

HOLBAEK, Denmark

WMO: 061560

Lat: 55.7358N

Lon: 11.6033E

Elev: 43

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	21.4	12.9	10.6	20.8	16.4	12.6	23.6	44.9	42.6	35.1	43.1	5.4	70	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	80.6	66.1	77.4	64.9	74.4	63.7	68.3	76.5	66.6	74.4	65.1	72.0	11.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	94.0	71.6	63.8	88.7	70.0	62.2	83.7	68.3	32.6	76.7	31.2	74.4	30.0	72.5	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.1	25.4	22.0	9.9	84.9	9.0	3.2	3.4	87.2	-1.9	89.0	-6.9	90.8	-13.5	93.1		
(5)			9.5	71.3	8.7	1.6	3.2	72.5	-1.8	73.4	-6.7	74.3	-13.0	75.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	33.5	34.1	38.2	46.0	53.6	58.6	63.8	63.5	57.4	49.4	42.4	36.7		
	DBStd	12.16	6.89	6.38	6.29	5.66	5.69	4.63	4.95	4.51	4.46	5.58	5.49	7.00		
	HDD50	2220	510	446	368	146	26	1	0	0	3	78	231	411		
	HDD65	6230	975	866	832	569	355	198	85	84	231	482	677	876		
	CDD50	1561	0	0	1	26	138	259	428	420	224	60	5	0		
	CDD65	97	0	0	0	0	2	6	47	39	3	0	0	0		
	CDH74	706	0	0	0	1	17	68	415	194	11	0	0	0		
	CDH80	101	0	0	0	0	1	4	71	25	0	0	0	0		
(14) Wind	WSAvg	10.4	12.2	11.9	11.4	10.5	9.6	9.8	8.6	8.9	9.4	10.4	11.2	11.4		
(15) Precipitation	PrecAvg	24.7	2.0	1.5	1.4	1.3	1.6	2.4	2.4	2.8	2.4	2.5	2.2	2.2		
	PrecMax	30.9	3.5	3.2	3.1	2.7	3.2	5.1	5.5	6.0	6.7	5.2	4.3	4.0		
	PrecMin	16.7	0.2	0.2	0.4	0.3	0.6	0.0	0.2	0.8	0.6	1.0	0.4	0.7		
	PrecStd	3.7	0.9	0.6	0.7	0.6	0.7	1.3	1.2	1.5	1.5	0.9	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	49.8	57.3	68.1	76.4	79.6	85.3	83.5	75.6	66.4	56.6	51.8		
		MCWB	47.7	46.3	49.2	55.9	64.0	65.3	68.4	67.8	64.0	59.9	54.7	49.8		
	2%	DB	46.6	47.1	52.6	63.8	71.8	75.7	81.9	79.2	71.4	61.2	54.0	49.7		
		MCWB	45.0	44.7	47.1	53.4	59.7	64.3	66.7	65.3	62.3	57.3	52.2	48.0		
	5%	DB	44.7	44.9	48.9	59.7	68.0	72.1	79.0	76.0	67.8	59.0	51.9	47.3		
		MCWB	43.0	42.8	44.8	50.9	58.4	62.2	66.0	64.3	60.6	56.0	50.3	45.8		
	10%	DB	42.8	42.9	46.6	56.1	64.7	68.6	75.4	72.9	65.0	57.0	49.9	45.4		
		MCWB	41.3	41.1	43.5	49.2	56.2	60.2	64.5	63.3	59.0	54.3	48.2	44.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.1	47.5	50.6	57.6	65.9	68.4	71.1	70.0	65.4	61.0	55.1	50.4		
		MCDB	49.5	49.0	54.8	66.4	73.8	76.3	81.3	77.9	72.0	64.3	56.3	51.5		
	2%	WB	45.3	44.9	47.5	54.5	62.0	65.6	68.9	67.7	63.2	58.1	52.5	48.2		
		MCDB	46.4	46.6	51.2	62.1	68.6	73.1	77.9	74.8	68.9	60.5	53.7	49.5		
	5%	WB	43.2	43.1	45.6	52.1	59.4	63.5	67.3	66.0	61.6	56.3	50.3	45.9		
		MCDB	44.4	44.7	48.4	57.9	65.7	70.1	75.8	72.7	66.6	58.4	51.6	47.1		
	10%	WB	41.4	41.1	43.8	49.8	57.2	61.2	65.6	64.5	60.0	54.7	48.5	44.2		
		MCDB	42.7	42.8	46.2	55.0	63.6	67.0	73.2	71.0	64.1	56.7	49.7	45.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.0	10.4	14.2	15.7	15.7	16.9	15.0	12.6	9.8	7.1	6.7		
		MCDBR	6.8	8.7	14.2	20.6	20.7	22.0	22.9	19.1	17.1	11.3	7.8	7.3		
	5% WB	MCWBR	6.6	7.2	10.1	12.2	12.2	12.3	10.5	8.7	9.5	8.3	7.2	7.1		
		MCWBR	7.0	8.6	13.4	17.9	18.4	19.6	20.5	17.1	15.6	10.8	7.5	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.337	0.361	0.395	0.389	0.383	0.405	0.405	0.376	0.356	0.337	0.304		
		taud	2.285	2.310	2.307	2.264	2.349	2.394	2.351	2.357	2.421	2.409	2.329	2.256		
	Ebn at Noon		187	230	254	261	270	273	264	256	247	221	177	160		
		Edh at Noon	17	24	31	37	36	35	36	34	28	22	17	14		
(44) All-Sky Solar Radiation	RadAvg		141	330	785	1312	1692	1820	1687	1345	951	471	181	100		
	RadStd		25	53	75	155	159	135	181	94	94	61	29	10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (67 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	-216	-311	N/A	N/A

Nomenclature: See separate page